PRESSURE PLAYER RANKINGS

A Data-Driven Analysis of Batting and Bowling Performance in T20 Cricket | 2022–2026

RESEARCH REPORT

Identifying players who performed above role-based expectations in selected high- and medium-pressure situations, and bowlers who recorded wickets at higher adjusted rates after opposition wickets had fallen.

Scope: T20 ball-by-ball dataset covering the 2022–2026 analysis window, including IPL and T20I records as represented in the supplied dataset.

Report basis: Verified outputs and rankings supplied during the analysis workflow. This report does not independently reprocess the source files.

Prepared as a reproducible analytical summary

1. Executive Summary

This project develops descriptive pressure-performance indicators for T20 batting and bowling. Batting is evaluated by comparing a batter's innings runs with a leave-one-out expected-runs baseline for the same batting position and entry phase. Bowling is evaluated using bowler-credited wickets per 100 recorded deliveries when the batting side had already lost a defined number of wickets, followed by a category-specific baseline adjustment.

The analysis distinguishes two situations: High pressure means the batting side had already lost 6–9 wickets before the recorded delivery or batter's first recorded appearance; Medium pressure means 3–5 wickets had been lost. These are operational definitions based on wickets lost, not direct measures of match state, required run rate, target, or win probability.

High-pressure batting	Romario Shepherd	+8.23 runs above expected per innings	25 innings
Medium-pressure batting	Towhid Hridoy	+10.88 runs above expected per innings	10 innings
High-pressure bowling	Jason Holder	11.12 adjusted wickets / 100 recorded deliveries	41 innings; 277 deliveries
Medium-pressure bowling	Mark Adair	7.15 adjusted wickets / 100 recorded deliveries	28 innings; 209 deliveries

The strongest robustness result is in high-pressure bowling: the same Top 10 membership appeared under prior strengths of 50, 100 and 200 deliveries, with only a small ordering change between two positions. Medium-pressure bowling showed more movement in its leading positions. Batting rankings are descriptive and should be interpreted alongside innings counts, medians and positive-innings percentages.

The findings should be described as pressure-performance proxies. The available data does not support claims that these players are definitively the most clutch, caused wins, or outperformed in verified match-winning situations.

2. Dataset and Data Quality

The analysis used ball-by-ball records in the supplied dataset for the 2022–2026 project window. The working dataset included IPL and T20I records. The following audit figures were established during the analysis:

Complete ball-by-ball records	192,465 recorded deliveries
Matches represented in complete-ball data	815 matches
First-innings recorded deliveries	100,531
Second-innings recorded deliveries	91,934
Matches with fewer than two innings	22; treated as incomplete/no-result cases for normal innings analysis
Matches with no ball-by-ball records	16
Duplicate rows in complete-ball table	0
Unique match–innings ball sequence check	innings_ball_num was unique within match and innings
Super Over records	26 deliveries across innings 3/4; excluded from normal innings analysis

The complete-ball table contained batter runs, four/six indicators, wicket flags, dismissal types, player IDs, batting order, over identifiers, and extras categories. Player IDs were retained as the primary key because a small number of names appeared with multiple IDs (including Abdul Samad and Haider Ali).

Important scoring limitation

The `runs` field represents batter runs rather than a fully reconstructed team total. Extras categories were available, but actual delivery-level extra-run amounts could not be reliably reconstructed from the supplied fields. A match-level scorecard cross-check illustrated why this matters: batter runs and extras together produced innings totals that differed from batter-run sums. Therefore, this report does not claim exact team scoring progression, official economy rates, official dot-ball rates, or official balls-faced calculations from this dataset.

The `overid` field was used for phase grouping. `over_ball_num` was not treated as a legal-ball count because it can include illegal deliveries and values beyond six within an over.

3. Methodology

3.1 Pressure definitions

Low	0–2 wickets lost before the relevant observation
Medium	3–5 wickets lost before the relevant observation
High	6–9 wickets lost before the relevant observation

For batting, wickets lost were counted before the batter's first recorded appearance as striker or non-striker in that innings. First recorded appearance is an approximation of crease-entry timing; it is not a verified timestamp of physical entry. For bowling, wickets before a delivery were calculated from all dismissals already recorded in the innings. Bowler-credit wickets were counted separately.

3.2 Batting metric

Expected runs were estimated from other innings in the same batting-order and entry-phase group, using a leave-one-out mean. Entry phases were Powerplay (overs 1–6), Middle (7–15) and Death (16–20). A minimum group sample size of 30 was required for the expected-runs comparison.

Runs above expected = batter's innings runs − leave-one-out mean runs for the same batting order and entry phase.

Player-level batting summaries report innings, total runs, average and median runs above expected, number of positive innings, and positive-innings percentage. Final CSV rankings included players with at least 10 innings in the respective pressure category.

This adjustment controls descriptively for batting position and entry phase. It does not isolate a causal pressure effect. Batting order and wickets lost before entry are closely related, so they should not be treated as independent pressure factors or double-counted in a composite score.

3.3 Bowling metric

Bowler-credit dismissals included Caught, Bowled, LBW, Stump and Hit Wicket. Run Out, Retired Out and Obstructing the Field were not credited to the bowler. The raw rate was calculated as bowler-credit wickets divided by recorded deliveries, multiplied by 100.

Raw wickets per 100 recorded deliveries = credited wickets ÷ recorded deliveries × 100.

Final bowling eligibility required at least 20 innings and 200 recorded deliveries in the relevant pressure category. Because recorded deliveries may include illegal deliveries, this metric is not an official bowling strike rate.

3.4 Baseline adjustment (shrinkage)

To reduce the influence of smaller samples, each bowler's raw rate was adjusted toward the overall rate within the same pressure category. The primary prior strength was 100 deliveries.

Adjusted rate = (player wickets + category benchmark × prior deliveries) ÷ (player deliveries + prior deliveries) × 100.

The category benchmarks were 7.93 wickets per 100 recorded deliveries for High pressure and 5.08 for Medium pressure. Prior strengths of 50, 100 and 200 deliveries were tested. This is a transparent heuristic shrinkage adjustment, not a formal Bayesian posterior or confidence interval.

4. Batting Results

4.1 High-pressure batting — Top 10

1	Romario Shepherd	25	462	8.23	8.12	68.0
2	Mahendra Singh Dhoni	17	267	7.59	5.02	64.71
3	Jamie Overton	10	155	5.96	4.57	60.0
4	Corbin Bosch	11	164	5.6	4.02	63.64
5	Mark Adair	18	257	4.32	1.52	55.56
6	Deepak Chahar	11	94	2.65	-3.66	45.45
7	Faheem Ashraf	21	245	2.54	0.1	57.14
8	Rashid Khan	49	566	2.52	-2.99	38.78
9	Mitchell Santner	15	164	2.51	-4.03	40.0
10	Harshit Rana	13	105	2.42	-0.99	38.46

Romario Shepherd leads the High-pressure batting table with +8.23 average runs above expected across 25 innings, with 68.0% positive innings. Dhoni ranks second (+7.59) from 17 innings. The smaller samples for Jamie Overton and Corbin Bosch warrant caution. Median performance and positive-innings percentage provide useful context: a high mean does not guarantee consistently positive outcomes.

4.2 Medium-pressure batting — Top 10

1	Towhid Hridoy	10	321	10.88	3.34	50.0
2	Heinrich Klaasen	44	1323	9.92	7.51	65.91
3	Nicholas Pooran	43	1184	7.81	-1.5	48.84
4	Matthew Wade	19	431	7.62	10.22	73.68
5	Tim David	81	1812	6.04	3.5	55.56
6	Rinku Singh	75	1761	6.03	1.76	53.33
7	David Miller	68	1637	5.7	0.79	54.41
8	Nitish Rana	10	268	5.35	-3.44	40.0
9	Shashank Singh	35	728	5.13	0.08	51.43
10	Moeen Ali	30	638	5.1	4.08	53.33

Towhid Hridoy has the highest mean (+10.88) but only 10 qualifying innings, so the estimate is less stable than the records for Klaasen, Pooran, Tim David, Rinku Singh and David Miller. Heinrich Klaasen combines a high average (+9.92) with 44 innings and a 65.91% positive-innings rate, making his result particularly notable within this table.

Batting interpretation note: the rankings compare innings runs with a role-and-entry-phase baseline. They do not account for exact target, required run rate, match result, partnership context, or verified team score progression.

5. Bowling Results

5.1 High-pressure bowling — Top 10

1	Jason Holder	41	277	34	12.27	11.12
2	Kuldeep Yadav	29	217	26	11.98	10.7
3	Shaheen Afridi	26	204	22	10.78	9.84
4	Mustafizur Rahman	34	314	31	9.87	9.4
5	Rashid Khan	44	379	35	9.23	8.96
6	Jasprit Bumrah	34	296	26	8.78	8.57
7	Harshal Patel	38	367	31	8.45	8.34
8	Taskin Ahmed	23	200	17	8.5	8.31
9	Arshdeep Singh	70	534	44	8.24	8.19
10	Varun Chakravarthy	37	281	23	8.19	8.12

Jason Holder leads with 34 credited wickets in 277 recorded deliveries across 41 innings. His adjusted rate is 11.12 per 100 recorded deliveries. Kuldeep Yadav and Shaheen Afridi rank second and third. Arshdeep Singh has the largest delivery sample among the listed players (534), which adds useful exposure context to his adjusted rate.

5.2 Medium-pressure bowling — Top 10

1	Mark Adair	28	209	17	8.13	7.15
2	Wanindu Hasaranga	70	841	62	7.37	7.13
3	Abrar Ahmed	26	266	21	7.89	7.13
4	Mohit Sharma	26	317	24	7.57	6.97
5	Khaleel Ahmed	37	321	24	7.48	6.91
6	Josh Hazlewood	50	401	29	7.23	6.8
7	Mohammad Nawaz	41	406	29	7.14	6.73
8	Varun Chakravarthy	77	891	61	6.85	6.67
9	Dasun Shanaka	23	221	16	7.24	6.57
10	Mujeeb Ur Rahman	29	254	18	7.09	6.52

Mark Adair leads the Medium-pressure adjusted ranking at 7.15 per 100 recorded deliveries, but his sample is 209 deliveries. Wanindu Hasaranga's 841 deliveries and 70 innings provide a much larger sample; his adjusted rate (7.13) is almost identical to Abrar Ahmed's (7.13) and close to Adair's. This illustrates why sample volume and adjusted rates should be read together.

6. Robustness and Sensitivity Analysis

6.1 Bowling prior-strength sensitivity

The rankings were recalculated using 50, 100 and 200 prior deliveries while keeping the eligibility threshold fixed at 20 innings and 200 deliveries. The Top 10 membership was unchanged in both High and Medium pressure across all three settings. The ordering was more stable in High pressure than in Medium pressure.

High	Holder, Kuldeep, Shaheen, Mustafizur, Rashid lead	Same leading five	Same leading five	Top five order unchanged; positions 7–8 swap
Medium	Adair, Abrar, Hasaranga lead	Adair, Hasaranga, Abrar lead	Hasaranga, Abrar, Adair lead	Top positions are prior-sensitive

In High pressure, Jason Holder remained first, followed by Kuldeep Yadav, Shaheen Afridi, Mustafizur Rahman and Rashid Khan for all tested prior strengths. In Medium pressure, Hasaranga rose from third at prior 50 to first at prior 200, while Adair moved from first to third. This is consistent with shrinkage giving more weight to the category benchmark as prior strength increases, particularly for smaller samples.

6.2 Batting sample sensitivity

High-pressure batting was also checked at higher minimum-innings thresholds during the analysis. Romario Shepherd remained the leading player at minimum thresholds of 10, 15 and 20 innings. At the 20-innings threshold, the eligible field narrowed substantially; several names in the 10- or 15-innings Top 10 no longer qualified. Therefore, the 10-innings batting tables in this report should be treated as broader exploratory rankings, not equally reliable estimates.

6.3 Interpretation

Robustness checks show that some leaders are not merely artifacts of one selected prior. However, stability under these specific thresholds does not prove that the metric measures true clutch ability. Further validation would require additional seasons, independent datasets, match-state variables and out-of-sample testing.

7. Key Findings

1. Romario Shepherd tops the High-pressure batting table, with +8.23 average runs above the role-based expected value across 25 innings.

2. Heinrich Klaasen is a notable Medium-pressure batter: +9.92 average runs above expected across 44 innings, with 65.91% positive innings.

3. Jason Holder leads the adjusted High-pressure bowling table at 11.12 wickets per 100 recorded deliveries, based on 277 deliveries across 41 innings.

4. Mark Adair leads the adjusted Medium-pressure bowling table at 7.15 per 100, but Hasaranga's much larger sample and near-identical adjusted rate make the leading positions close.

5. High-pressure bowling's leading order is more robust to prior-strength changes than Medium-pressure bowling's leading order.

6. Role and opportunity matter. Lower-order batters tend to enter after more wickets have fallen, so wickets-before-entry is partly a proxy for batting role and remaining innings opportunity.

7. Averages alone are insufficient. Median runs above expected, positive-innings percentage, innings count and delivery count should accompany every player ranking.

8. The available dataset supports comparative descriptive analysis, but not definitive match-winning or causal claims.

8. Limitations and Risks

No reliable delivery-level actual extras amounts	Cannot reconstruct exact team totals and score progression from the working fields.
No verified target / required run rate / win probability	Pressure is defined by wickets lost only; it is not full match-state pressure.
First recorded appearance is an entry proxy	May differ from the exact moment a batter physically entered the field.
Recorded deliveries may include illegal balls	Bowling rate is not official strike rate; do not infer official economy.
Batter-run field excludes complete extras	Do not interpret batter_runs as total runs conceded by bowler or team.
Small samples in some batting groups	High average values can be volatile; show innings and medians.
Role and pressure are correlated	Cannot infer pressure caused the observed performance difference.
Dataset scope and coverage	Results apply only to the supplied 2022–2026 records and their coverage.

9. Conclusion

This analysis provides a transparent first version of a T20 pressure-performance framework. It compares batting output with a role- and entry-phase baseline, and compares bowler wicket frequency within wicket-loss categories using category-specific shrinkage. The strongest descriptive signals in the supplied results are Romario Shepherd in High-pressure batting, Heinrich Klaasen in Medium-pressure batting, Jason Holder in High-pressure bowling and Mark Adair in Medium-pressure bowling.

The project should present these as leaders under the chosen operational definitions and eligibility rules—not as definitive all-time clutch rankings. High-pressure bowling shows encouraging ranking stability across the tested prior strengths. Medium-pressure bowling is more sensitive, and some batting leaders have modest sample sizes.

10. Recommended Next Research Steps

1. Obtain delivery-level total runs, legal-ball flags, no-ball/wide run amounts and complete innings totals from a scorecard-validated source.

2. Add target, runs required, balls remaining, required run rate, wickets in hand, innings number, match stage and match result to construct richer pressure states.

3. Separate batting and bowling formats, leagues and international competitions where sample sizes permit; test whether rankings transfer across competitions.

4. Use out-of-sample validation: build expected-performance baselines on earlier seasons and evaluate later seasons.

5. Add uncertainty intervals or bootstrap resampling for player rates and batting above-expected estimates.

6. Test alternative definitions of pressure and minimum sample thresholds before presenting a single composite ranking.

11. Metric Reference

Average runs above expected	Mean innings runs minus leave-one-out mean for same batting order and entry phase.
Median runs above expected	Median of the same innings-level difference; reduces sensitivity to extreme innings.
Positive innings %	Share of qualifying innings where runs above expected is greater than zero.
Raw wickets / 100	Credited bowler wickets per 100 recorded deliveries in the pressure category.
Adjusted wickets / 100	Raw wicket rate shrunk toward the category benchmark using the selected prior strength.

End of report